

CSL1010: Introduction to Computer Science

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Time: 120 Minutes

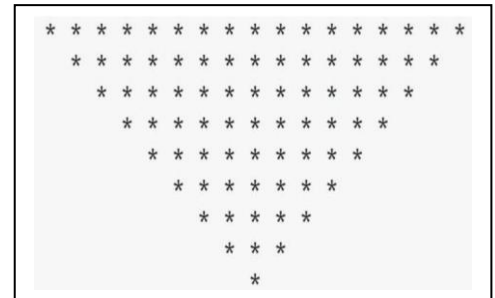
Full Marks: 50

Attempt all questions. Use the answer sheet for rough work and write your final answers neatly and clearly in the space provided on the question paper. Read each question carefully before attempting. **Good Luck!**

1. Fill in the blanks to produce the pattern shown on the right.

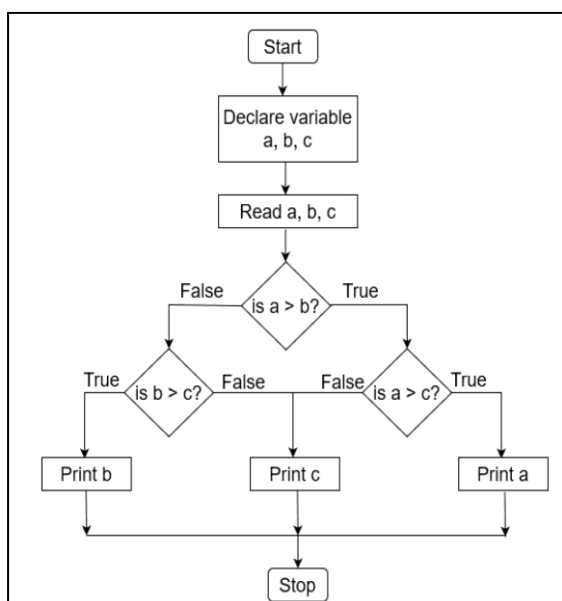
5 * 2

```
void main ()
{
    int i, j, rows, k, m = 0;
    printf (" Enter a number to define the rows: \n");
    scanf ("%d", &rows);
    printf("\n");
    for (i = __rows__; i >= 1; i--)
    {
        for (j = 1; j <= __m__; j = j + _1_)
        {
            printf (" "); // print the space
        }
        for (k = 1; k <= __(2 * i - 1__); k++)
        {
            printf ("* "); // print the Star
        }
        m++;
        __printf("\n");__
    }
}
```



2. Study the flow diagram below and write the code in the space provided on the right.

10



```
#include <stdio.h>
```

```
int main() {
    int a, b, c, max;
```

```
    // Input three numbers
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
```

```
    // Compare and find maximum
    if (a >= b && a >= c)
        max = a;
    else if (b >= a && b >= c)
        max = b;
    else
        max = c;
```

```
    // Output result
    printf("The maximum number is: %d\n", max);
```

```
    return 0;
}
```

Name:

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Department:

3. Match the items on the left-hand side with their appropriate pairs on the right-hand side. 5 * 2

1. Preprocessing	1. Loads the executable into memory and assigns runtime addresses.
2. Compilation	2. Facilitates the transition from runtime initialization to the user entry point.
3. <code>_libc_start_main()</code>	3. Resolves symbolic references and merges multiple object files into one unit.
4. Linking	4. Expands macros, removes comments, and inserts contents of header files.
5. Loader	5. Generates human-readable low-level instructions specific to the target CPU.

Answer:

1 → 4

2 → 5

3 → 2

4 → 3

5 → 1

4. Fill in the blanks to complete the program that finds the Greatest Common Divisor (GCD) of two positive integers entered by the user. 5 * 2

Example of GCD:

- Divisors of 36 → 1, 2, 3, 4, 6, 9, 12, 18, 36
- Divisors of 60 → 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60
- Common divisors → 1, 2, 3, 4, 6, 12
- **Greatest common divisor (gcd) = 12**

```
#include <stdio.h>
```

```
int main() {
```

```
    int a, b, i, gcd = 1;
```

```
    printf("Enter two numbers: ");
```

```
    scanf("%d %d", &a, &b);
```

```
    // Find the smaller number
```

```
    int min = (a < b) ? a : b
```

```
    for (i = 1; i <= min; i++) {
```

```
        if (a % i == 0 && b % i == 0) {
```

```
            gcd = i;
```

```
        }
```

```
    }
```

```
    printf("GCD = %d\n", gcd);
```

```
    return 0;
```

```
}
```

Name:

Roll:

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5. Find the output of each of the following C programs.

5 * 2

a.

```
int main() {
    int i, j, k;
    for (i = 1; i <= 3; i++) {
        for (j = 1; j <= i; j++) {
            for (k = 1; k <= j; k++) {
                printf("%d", k);
            }
            printf(" ");
        }
        printf("\n");
    }
}
```

Answer:

```
1
1 12
1 12 123
```

b.

```
int main() {
    int i = 1, j;
    while (i <= 3) {
        j = 1;
        while (j <= i) {
            printf("%d", i + j);
            j++;
        }
        printf("\n");
        i++;
    }
}
```

Answer:

```
2
34
456
```

c.

```
int main() {
    int i = 5;
    while(i = 0) {
        printf("%d ", i);
        i--;
    }
}
```

Answer: nothing will be printed

d.

```
int main() {
    int i, j;
    for(i = 1; i <= 3; i++) {
        for(j = 1; j <= 3; j++) {
            if(i + j == 4)
                printf("# ");
            else
                printf("* ");
        }
        printf("\n");
    }
}
```

Answer:

```
* * #
* # *
# * *
```

e.

```
int main() {
    int i = 0;
    for(i = 0; i < 3; )
        printf("%d ", i++<<3); }
```

Answer: 0 8 16